

GENERAL SPECIFICATION G4.62

Metal Tanks

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G4.62 METAL TANKS

This Specification defines mandatory requirements for the design, materials, fabrication, supply and assembly of field erected welded and field erected bolted metal tanks.

G4.62.1 Field Erected Welded Water Tanks

(a) Scope of Supply

The Contractor shall furnish the following scope of services as part of the work under this specification:

- (i) All materials, equipment and accessories as defined herein;
- (ii) Drawings and design documentation certified by a Professional Engineer (P.E.) registered in the appropriate jurisdiction;
- (iii) Delivery and erection of the welded water tanks;
- (iv) Tests as defined herein; and
- (v) Instruction manuals.

(b) Items furnished by others and interfaces

- (i) Items furnished by others and not in this scope of supply include the following:
 - 1) Concrete tank foundation(s) as specified on the Field Erected Welded Water Tanks Specification Sheet(s) which may not be available at the time of release of this specification.

(c) Performance and Design Requirements

- (i) Performance and design requirements are covered in the Field Erected Welded Tanks Specification Sheet(s).

(d) Codes and Standards

- (i) Work performed under these specifications shall be done in accordance with the following codes and standards described herein.
- (ii) Unless otherwise specified, the applicable governing edition and addenda to be used for all references to codes or standards specified herein shall be interpreted to be the latest available jurisdictionally approved edition and addenda.
- (iii) If a code or standard is not jurisdictionally mandated, then the current edition and addenda in effect at the date of this document shall apply. These references shall govern the work except where they conflict with the specifications. In case of any conflict, the latter shall govern to the extent of such difference:

Work	In Accordance With
Design	AWWA D100 or API 650; NFPA where applicable
Grounded pad screw hole spacing and location	NEMA
Tank erection, testing, and coating	AWWA D100 or API 650
Cleanliness inspection before painting	SSPC Visual Standard, SSPC VIS 1
Disinfection of tank interior	AWWA C652
Pipe flanges	BS EN 1092/ASME/ANSI B16.5
Forged steel pipe fittings	ANSI B16.11
Pipe/flange butt weld end preparation details	ANSI B16.25
Proportions for mixing of grout filler	AWWA D100
Use of cane-fiber joint filler for tank support	AWWA D100

(e) Materials

The following materials shall be used:

Component	Material
Shell plate, pipe flange covers, and manhole necks (if not seamless)	ASTM A36 or ASTM A516 Grade 60
Nozzles or seamless manhole necks	ASTM A106 Grade B
Structural shapes and plates	ASTM A36
Forged steel flanges	ASTM A105
Grounding pads	ASTM A167, Type 304
Overflow piping	ASTM A53, Std Wt
Appurtenance flange bolting and nuts	ASTM A307 Gr A or ASTM A325 Type 1 bolts with hex heads o studs; ASTM A563 Gr A hex nuts; all galvanized
Gaskets	Buna N (Nitrile)
Cement	ASTM C150, Type I
Sand	ASTM C33
Cane-fiber joint filler	ASTM D1751
Tank roof walkways	Serrated grating
Two-component polysulfide rubber sealant	Fed Spec TT-S-00227E, Type II, Class A
Cement	ASTM C150, Type 1
Sand	ASTM C33

(f) Test Requirements

- (i) The following testing shall be conducted in accordance with the specified source. This testing is to be considered part of the defined Scope of Work, and all associated costs shall be the responsibility of the Contractor unless specifically identified otherwise.
- (ii) The Contractor is responsible for all costs associated with correcting deficiencies and retesting in the event of a test failure:

Tests	In Accordance With	Conducted By
All weld tests	AWWA D100 or API 650	Contractor/Contractor's appointed representative
Leak tests	Refer to requirements on Testing specified herein	Contractor/Contractor's appointed representative

(g) Products

- (i) Each tank shall be a vertical, cylindrical, above-ground, atmospheric water storage tank;
- (ii) Each tank shall be designed for field erection by welding; and
- (iii) Each tank shall be designed to be set upon a Type 1 or Type 2 foundation as specified in the requirements for foundation later in this specification.

(h) Design Criteria

- (i) The thicknesses of the tank roof, shell, and bottom plates shall be as required by the specified design code(s) plus corrosion allowance(s) if specified on the Field Erected Welded Water Tanks Specification Sheet(s); and
- (ii) Design pressure for all tanks shall be atmospheric pressure.

(i) Material Test Reports

- (i) Certified mill test reports shall be furnished covering all steel plates and structural shapes to be used in the work as evidence that such materials are new and comply with the referenced standards.

(j) Fabrication Requirements

- (i) Tank sections shall be shop fabricated to the maximum extent practical to minimize field welding;
- (ii) A self-supporting roof shall be provided;
- (iii) A crowned tank bottom shall be provided with slope of 1:120;
- (iv) Shell plate joints shall be double-welded butt joint type welds with complete penetration;
- (v) Roof joints shall be complete penetration welds or lap welds welded continuously on both edges;
- (vi) Bottom plates shall be complete penetration welds or lap welded on top side;
- (vii) All field joints shall be properly identified on the erection drawings. Joints shall be acceptable to the S.O. and approved welding inspectors; and
- (viii) Welding shall be performed in accordance with the specified design codes.

- (k) Tank Appurtenances
 - (i) Specified tank appurtenances shall be furnished; and
 - (ii) Exterior appurtenances shall be designed for the wind and seismic loads as far as applicable to the site conditions.
- (l) Roof Manholes
 - (i) Two roof manholes shall be provided on each tank, each with an inside diameter of 610 mm; and
 - (ii) Each manhole shall have a hinged, rainproof cover with locking device.
- (m) Shell Manholes
 - (i) Two shell manholes shall be provided on each tank; and
 - (ii) Shell manholes shall be 610 mm diameter hinged closures.
- (n) Manhole Gaskets
 - (i) Two sets of permanent gaskets shall be furnished for all manhole covers; and
 - (ii) The permanent gaskets shall be furnished in wooden boxes identified by tank name.
- (o) Piping Connections
 - (i) Piping nozzles shall be provided as required according to the tender P and IDs;
 - (ii) Flanges shall be raised face to BS EN 1092, rated PN10 as a minimum and weld neck type;
 - (iii) Unless otherwise indicated on the drawings, tank connections 50 mm and smaller shall be ANSI Pressure Class 3000 integrally reinforced forged branch outlet fittings; and
 - (iv) Each tank outlet that is larger than 100 mm in diameter shall be provided with an anti-vortexing device.
- (p) Overflow
 - (i) An external overflow discharge pipe shall be provided to convey overflow of the tank to near ground level. It shall be sized to handle the maximum tank inflow rate as specified.
 - (ii) The discharge end of the overflow pipe shall be provided with an elbow and flap valve.
 - (iii) Overflow piping shall be rigidly supported and braced from the tank wall at intervals not exceeding 1,800 mm.
 - (iv) The piping shall be adequately supported and braced for weight, expansion or contraction, wind and hydraulic forces. U-bolts for support members shall not be used. If required, tank shell wall thickness can be supplemented by providing spreading pads on the shell where pipe supports are attached.

- (q) Filter and Breather Valve
 - (i) If specified, a corrosion-resistant filter and a breather valve shall be provided for each tank; and
 - (ii) The valve shall be sized for the maximum flow rates applicable.
- (r) Grounding Pads
 - (i) Four stainless steel grounding pads shall be provided, equally spaced around the exterior of the tank and welded to the tank shell 150 mm above the bottom.
 - (ii) Pads shall be provided that are 50 mm by 90 mm by 13 mm thick, drilled and tapped to full thickness for M10 x 1.5 hex head cap screws.
 - (iii) Screw hole spacing and location shall be matched to that of NEMA standard two-hole terminals.
 - (iv) Flat and smooth contact surfaces shall be provided and protected during handling to maintain contact surface quality.
- (s) Shop Coating
 - (i) Exterior surfaces of tank roof, shell plates and tank bottom shall be shop primed where so indicated.
 - (ii) Plate preparation and coating shall be in accordance with the General Specification on Painting and Protection.
- (t) Access Provisions

Access provisions shall be in accordance with the General Specification on Metalwork and Workmanship Requirements.
- (u) Exterior Ladder

Where specified or indicated on the drawings, an exterior, caged ladder shall be provided. Gates shall be provided for ladders in accordance with OSHA requirements and the General Specification on Metalwork and Workmanship Requirements.
- (v) Interior Ladder

Where specified or indicated on the drawings, an interior ladder shall be provided with a fall prevention ladder climbing system.
- (w) Exterior Stairs
 - (i) Where specified or indicated on the drawings, exterior stairs shall be provided with a top landing platform and intermediate landing platform(s);
 - (ii) The stairs shall extend from the bottom to the top of the tank and shall be spiral type;
 - (iii) Each stair landing platform shall incorporate the best access and most economical use of space in platform design;
 - (iv) Where specified or indicated on the drawings, a common platform shall be provided spanning the gap between two tanks; and

- (v) The platform attachment shall be designed so that movement of either tank resulting from expansion, contraction, or settlement will not impose undue forces on either tank or on the platform structure.
- (x) Walkway
 - (i) A 900 mm wide walkway shall be provided on each tank roof;
 - (ii) The walkway shall extend from the upper ladder termination or stair landing platform to the manholes and other roof mounted appurtenances; and
 - (iii) The walkway shall be completely enclosed with handrails and kick plates except at the ladder or landing platform access opening.
- (y) Handling
 - (i) Shop fabricated tank sections and structural components shall be handled during shipping, storage, and erection in a manner that will prevent warping, bending, distortion or physical damage of any kind.
- (z) Tank Foundation
 - (i) The tank support concrete foundation(s) shall comply to a separate Civil and Structural General Specification, unless otherwise specified.
 - (ii) For Type 1 (concrete ring wall with soil infill) foundations, a layer of clean sand, finely crushed limestone, or clam shells compacted to a minimum thickness of 76 mm shall be furnished and installed by the Contractor between the earthen interior of the ring wall and the tank bottom. The clean fill material shall be kept free of topsoil, clay, organic material, and other deleterious substances. A 25 mm minimum space between the tank bottom and top of the ring wall shall be grouted full with portland cement grout. Alternatively, 13 mm thick cane-fibre joint filler in accordance with AWWA D100 may be used in lieu of the 25 mm of grout.
 - (iii) For Type 2 (concrete mat or concrete ring wall with concrete infill) foundations, a minimum 13 mm thick cane-fibre joint filler meeting the requirements of ASTM D1751 shall be furnished and installed by the Contractor between the tank bottom and concrete slab. The tank shell shall be supported with Portland cement grout. Alternatively, 13 mm thick cane-fibre joint filler in accordance with AWWA D100 may be used.
 - (iv) When portland cement grout is used as the bearing surface for the tank shell, it shall be installed after erection of the first shell course. Sand and water used for grouting the tank bottom shall meet the following minimum requirements:
 - 1) Sand-Fine aggregate, clean, well graded, natural sand; and
 - 2) Water-Clean and free from mud, oil, organic matter, and other deleterious substances.
 - (v) Grout shall be mixed and placed in accordance with the manufacturer's recommendations and in the proportions specified in AWWA D100. Only enough water shall be added to produce a grout that can be tamped in place. All grout shall be thoroughly mixed for not less than 5 minutes in a mechanical mixer of an approved type. After mixing, the grout shall be used immediately and before stiffening sets in.

- (vi) After complete curing of the grout, the peripheral joint between the tank bottom plate and the foundation shall be completely sealed as indicated on the seal standard details drawing supplementing this specification. A two-component polysulfide rubber sealant shall be applied in accordance with the sealant manufacturer's recommendations, including removal of oil from all surfaces in contact with the sealant.
- (aa) Tank Erection
 - (i) Welding shall be in accordance with the General Specification on Metalwork and Workmanship Requirements; and
 - (ii) Tank appurtenances shall be installed in accordance with referenced codes, standards and manufacturer's instructions.
- (bb) Surface Preparation, Testing, Painting, Cleaning, and Disinfecting of Water Tanks
 - (i) All clips and lugs used for erection shall be cut off. Any remaining clip and lug attachment weld material, and any weld used for fit-up shall be ground smooth and flush with the parent metal. All welds shall be ground to remove any roughness or irregularity. All weld spatter shall be removed.
 - (ii) On the interior of the tank, all butt welds with remaining weld bead crown height not exceeding 1.6 mm shall be ground smooth.
 - (iii) All sharp metal edges existing at fillet welds shall be ground to a minimum 3.2 mm radius.
 - (iv) All surfaces shall be cleaned of lubricants.
- (cc) Testing
 - (i) All testing and repair work shall be accepted by the S.O. after any weld grinding, but before blasting or painting is started.
 - (ii) Welded joints shall be radiographically inspected in the shell of the tank as required by the applicable referenced codes and standards.
 - (iii) Joints shall be tested by vacuum box method where required by the applicable referenced codes and standards.
 - (iv) Repair of defective welds and leaking joints shall be in accordance with the applicable referenced codes and standards.
 - (v) After completion of welding, but before beginning surface preparation for painting or coating, and before any external piping has been connected, the tank shall be filled with water and all submerged joints shall be inspected.
 - (vi) Any leaking joints shall be repaired after the water level has been dropped to at least 610 mm below the point being repaired. The test shall be repeated until all joints are completely leak tight (zero leakage and seepage).
- (dd) Painting
 - (i) Iron and steel surfaces of the tank and accessories shall be prepared and painted in accordance with the General Specification on Painting and Protection.

- (ee) Cleaning and Disinfecting
 - (i) After erection, testing and painting have been completed, the interior of the tank shall be thoroughly cleaned and all debris shall be removed.
 - (ii) If specified on the Field Erected Welded Water Tank Specification Sheet(s), the interior of the tank shall be disinfected in accordance with the referenced standards as well as state and local laws and ordinances, whichever requirement(s) is/are the most stringent. Prior to starting any cleaning and disinfecting work, the Contractor shall submit to the S.O. the detailed procedures proposed, including the coordination and sequence of operations.
 - (iii) The Contractor shall provide all cleaning and disinfecting materials, all equipment and labour necessary for the cleaning and disinfecting operations.
 - (iv) All wastewater from disinfecting the tank shall be disposed off in a manner acceptable to the S.O. and the appropriate pollution control agency.

G4.62.2 Field Erected Bolted Water Tanks

- (a) Scope of Supply
 - (i) The Contractor shall furnish the following scope of services as part of the work under this specification:
 - 1) All materials, equipment, and accessories as defined herein;
 - 2) Drawings and design documentation certified by a Professional Engineer (P.E.) registered in the appropriate jurisdiction;
 - 3) Delivery and erection of the bolted tanks;
 - 4) Tests as defined herein; and
 - 5) Instruction manuals.
- (b) Items Furnished by Others and Interfaces

Items furnished by others and not in this scope of supply include the following:

 - (i) Concrete tank foundation(s) as specified on the Field Erected Bolted Tanks Specification Sheet(s) which may not be available at the time of release of this specification.
- (c) Performance and Design Requirements
 - (i) Performance and design requirements are covered in the Field Erected Bolted Tanks Specification Sheet(s).
- (d) Codes and Standards
 - (i) Work performed under these specifications shall be done in accordance with the following codes and standards.
 - (ii) Unless otherwise specified, the applicable governing edition and addenda to be used for all references to codes or standards specified herein shall be interpreted to be the jurisdictionally approved edition and addenda.

- (iii) If a code or standard is not jurisdictionally mandated, then the current edition and addenda in effect at the date of this document shall apply. These references shall govern the work except where they conflict with the Board's specifications. In case of conflict, the latter shall govern to the extent of such difference:

Work	In Accordance With
Design	AWWA D103; NFPA where applicable
Grounded pad screw hole spacing and location	NEMA
Tank erection, testing, and coating	AWWA D103
Surface preparation before painting	SSPC
Disinfection of tank interior	AWWA C652
Pipe flanges	BS EN 1092/ASME/ANSI B16.5
Forged steel pipe fittings	ANSI B16.11

- (e) Materials

The following materials shall be used:

Component	Material
Shell plate, pipe flange covers, and manhole necks (if not seamless)	ASTM A36 or acceptable equal
Nozzles or seamless manhole necks	ASTM A106 Grade B
Structural shapes and plates	ASTM A36
Forged steel flanges	ASTM A105
Tank joint bolts	ASTM A307, ASTM A325, or API 12B galvanized in accordance with ASTM A153
Grounding pads	ASTM A167, Type 304
Overflow piping	ASTM A53, std wt
Appurtenance flange bolting and nuts	ASTM A307 Grade A or ASTM A325 Type 1 bolts with hex heads or studs; ASTM A563 Grade A hex nuts; all galvanized
Gaskets (manhole)	Buna N (Nitrile)
Gaskets (tank joints)	Manufacturer's standard compatible with the stored liquid
Two-component polysulfide rubber sealant	Fed Spec TT-S-00227E, Type II, Class A
Cement	ASTM C150, Type I
Sand	ASTM C33

(f) Test Requirements

- (i) The following testing shall be conducted in accordance with the specified source. This testing is to be considered part of the defined Scope of Work, and all associated costs are the responsibility of the Contractor unless specifically identified otherwise.
- (ii) The Contractor is responsible for all costs associated with correcting deficiencies and retesting in the event of a test failure:

Tests	In Accordance With	Conducted By
All weld tests	AWWA D103	Contractor/Contractor's appointed representative
Leak tests	Refer to requirements on Testing specified herein	Contractor/Contractor's appointed representative

(g) Products

- (i) Each tank shall be a vertical, cylindrical, aboveground, atmospheric water storage tank;
- (ii) Each tank shall be designed for field erection by bolting; and
- (iii) Each tank shall be designed to be set upon a Type 1 or Type 2 foundation as specified in the requirements for foundation later in this specification.

(h) Design Criteria

- (i) The thicknesses of the tank roof, shell and bottom plates shall be as required by AWWA D103 and NFPA, plus corrosion allowance(s) if specified on the Field Erected Bolted Tanks Specification Sheet(s); and
- (ii) Design pressure for all tanks shall be atmospheric pressure.

(i) Material Test Reports

- (i) Certified mill test reports, covering all steel plates and structural shapes to be used in the work shall be furnished as evidence that such materials are new and comply with the referenced standards.

(j) Fabrication Requirements

- (i) Tank sections shall be shop fabricated to the maximum extent practical to minimize field erection;
- (ii) The tank roof shall be a self-supporting type;
- (iii) The tank bottom shall be crowned, with a slope of 1:120;
- (iv) All shell, roof and bottom plate joints shall be of the bolted type designed in accordance with the requirements of AWWA D103;
- (v) All field joints shall be properly identified on the erection drawings. Joints shall be acceptable to the S.O. and approved welding inspectors; and
- (vi) Welding shall be performed in accordance with AWWA D103.

- (k) Tank Appurtenances
 - (i) Specified tank appurtenances shall be furnished; and
 - (ii) Exterior appurtenances shall be designed for the wind and seismic loads specified as far as applicable to the site of the project.
- (l) Roof manholes
 - (i) Two roof manholes shall be provided on each tank as indicated on the drawing(s), each with an inside diameter of 610 mm;
 - (ii) Each manhole shall have a hinged, rainproof cover with locking device;
 - (iii) Two shell manholes shall be provided on each tank as indicated on the drawing(s). Shell manholes shall be 610 mm diameter hinged closures; and
 - (iv) Two sets of permanent gaskets shall be furnished for all manhole covers. The permanent gaskets shall be furnished in wooden boxes identified by tank name.
- (m) Piping Connections
 - (i) Piping connections shall be provided as indicated on the drawings and as specified;
 - (ii) Flanges shall be raised face to BS EN1092, PN10 as a minimum, weld neck type flanges;
 - (iii) Butt welding ends shall be prepared in accordance with the detail drawings;
 - (iv) Unless otherwise indicated on the drawings, tank connections 50 mm and smaller shall be ANSI Pressure Class 3000 integrally reinforced forged branch outlet fittings; and
 - (v) Each tank outlet that is larger than 100 mm in diameter shall be provided with an anti-vortexing device.
- (n) Overflow
 - (i) An external overflow discharge pipe shall be provided to convey overflow of the tank to near ground level. It shall be sized to handle the maximum tank inflow rate as specified.
 - (ii) The discharge end of the overflow pipe shall be provided with an elbow and flap valve.
 - (iii) The overflow piping shall be rigidly supported and braced from the tank wall at intervals not exceeding 1,800 mm. The piping shall be adequately supported and braced for weight, expansion or contraction, wind, and hydraulic forces. U-bolts shall not be used for support members. If required, tank shell wall thickness can be supplemented by providing spreading pads on the shell where pipe supports are attached.
- (o) Filter and Breather Valve
 - (i) A corrosion-resistant filter and a breather valve for each tank shall be provided if indicated on the drawings or specified herein; and
 - (ii) The valve shall be sized for the maximum flow rates applicable.

- (p) Grounding Pads
 - (i) Four stainless steel grounding pads shall be equally spaced around the exterior of the tank and welded to the tank shell 150 mm above the bottom.
 - (ii) Pads 50 mm by 90 mm by 13 mm thick shall be drilled and tapped to full thickness for two M10 x 1.5 hex head cap screws.
 - (iii) Screw hole spacing and location shall be matched to that of NEMA standard two-hole terminals.
 - (iv) Flat and smooth contact surfaces shall be protected during handling to maintain contact surface quality.
- (q) Shop Coating
 - (i) Interior and exterior surfaces of tank bottom, roof and shell plates shall be finish coated in the manufacturer's shop.
- (r) Access Provisions
 - (i) Access provisions shall be in accordance with the General Specifications on Metalwork and Workmanship Requirements and the requirements herein:
- (s) Exterior Ladder
 - (i) Where specified or indicated on the drawings, an exterior caged ladder shall be provided.
- (t) Exterior Stairs
 - (i) Where specified or indicated on the drawings, exterior stairs with a top landing platform and intermediate landing platform(s) shall be provided;
 - (ii) The stairs shall extend from the bottom to the top of the tank and shall be spiral type;
 - (iii) Each stair landing platform shall incorporate the best access and most economical use of space in platform design; and
 - (iv) Where specified or indicated on the drawings, a common platform shall be provided to span the gap between two tanks. The platform attachment shall be designed so that movement of either tank resulting from expansion, contraction or settlement will not impose undue forces on either tank or on the platform structure.
- (u) Walkway
 - (i) A 900-mm wide non-skid walkway shall be provided on each tank roof;
 - (ii) The walkway shall extend from the upper ladder termination or stair landing platform to the manholes and other roof mounted appurtenances; and
 - (iii) The walkway shall be completely enclosed with handrails and kick plates except at the ladder or landing platform access opening.
- (v) Handling
 - (i) Shop fabricated tank sections and structural components shall be handled during shipping, storage and erection in a manner that will prevent warping, bending, distortion or physical damage of any kind.

(w) Tank Foundation

- (i) The tank support concrete foundation(s) shall comply to a separate Civil and Structural General Specification, unless otherwise specified.
- (ii) For Type 1 (concrete ring wall with soil infill) foundations, a layer of clean sand, finely crushed limestone, or clam shells compacted to a minimum thickness of 76 mm shall be furnished and installed by the Contractor between the earthen interior of the ring wall and the tank bottom. The clean fill material shall be kept free of topsoil, clay, organic material, and other deleterious substances. A 125 mm minimum space between the tank bottom and the top of the ring wall shall be grouted full with portland cement grout. Alternatively, a 13 mm thick cane-fibre joint filler in accordance with AWWA D103 may be used in lieu of the 25 mm of grout.
- (iii) For Type 2 (concrete mat or concrete ring wall with concrete infill) foundations, a minimum 13 mm thick cane-fibre joint filler meeting the requirements of ASTM D1751 shall be furnished and installed by the Contractor between the tank bottom and concrete slab. The tank shell shall be supported with portland cement grout or, alternatively, 13 mm thick cane-fibre joint filler in accordance with AWWA D103.
- (iv) When portland cement grout is used as the bearing surface for the tank shell, it shall be installed after erection of the first shell course. Sand and water used for grouting the tank bottom shall meet the following minimum requirements:
 - 1) Sand-Fine aggregate, clean, well graded, natural sand; and
 - 2) Water--Clean and free from mud, oil, organic matter, and other deleterious substances.
- (v) Grout shall be mixed and placed in accordance with the manufacturer's recommendations and in the proportions specified in AWWA D103. Only enough water shall be added to produce a grout that can be tamped in place. All grout shall be thoroughly mixed for not less than 5 minutes in a mechanical mixer of an approved type. After mixing, the grout shall be used immediately and before stiffening sets in.
- (vi) After complete curing of the grout, the peripheral joint between the tank bottom plate and the foundation shall be completely sealed as indicated on the seal standard details drawing supplementing this specification. A two-component polysulfide rubber sealant shall be applied in accordance with the sealant manufacturer's recommendations, including removal of oil from all surfaces in contact with the sealant.

(x) Tank Erection

- (i) Tank appurtenances shall be installed in accordance with referenced codes, standards and manufacturer's instructions.

(y) Testing

- (i) All testing shall be performed in accordance with AWWA D103 and as stated herein;
- (ii) Bottom joints shall be tested by the vacuum box method;

- (iii) All labour, materials and equipment including pumps, hoses and valves necessary for testing operations shall be provided; and
- (iv) Where deemed necessary, the Contractor shall be required to submit a full leak test procedure and methodology including timeline schedule for approval prior to conducting leak tests for the tanks.
- (z) Touch-up Painting
 - (i) Damaged coatings on any surface of the tank and accessories shall be repaired in accordance with the coating manufacturer's recommendation.
- (aa) Cleaning and Disinfecting
 - (i) After erection and testing have been completed, the tank shall be thoroughly cleaned and all debris shall be removed from the interior of the tank.
 - (ii) If specified, the tank interior shall be disinfected in accordance with the requirements of AWWA C652 as well as state and local laws and ordinances, whichever requirement(s) is/are the most stringent. Prior to starting any cleaning and disinfecting work, the Contractor shall submit to the S.O. the detailed procedures proposed including the coordination and sequence of operations.
 - (iii) All cleaning and disinfecting materials and all equipment and labour necessary for the cleaning and disinfecting operations shall be provided.
 - (iv) All wastewater from disinfecting the tank shall be disposed of in a manner acceptable to the S.O. and the appropriate pollution control agency.

G4.62.3 Field Erected Welded Screening and Grit Hopper

- (a) General
 - (i) The hoppers shall be of welded Type 316L stainless steel construction. Bolted and gasketed hopper construction will not be acceptable.
 - (ii) The hopper shall be structurally adequate to safely support its own weight, the full weight of material, and the associated live/dead loads of the appurtenances.
 - (iii) The hopper shall be designed and endorsed by a P.E. of the relevant discipline.
 - (iv) Hopper plates shall have a minimum thickness of 6 mm unless otherwise specified. The hoppers shall be watertight.
 - (v) Hopper shall be shop fabricated with structural reinforcement as required. Hopper discharge flanges and flange bolt holes shall be coordinated to match corbel locations and adjoining equipment locations.
 - (vi) The following items shall be included as part of the hopper fabrication. Verify all necessary dimensions and weights of connecting equipment before fabrication.
 - 1) Hopper Cone: Flanged for mounting discharge gate;
 - 2) Vent piping inlet and outlet;
 - 3) Manway with bolted, flush hinged cover;
 - 4) Stainless steel anchor bolts; and
 - 5) Weigh elements, transmitters and associated mounting and supports.

- (vii) Hopper interior shall be smooth, with no protruding flanges or structural members which could impede the flow of material. Hopper interior shall be lined with an aromatic polyurea elastomeric coating unless otherwise specified.
- (viii) Lifting Lugs shall be provided for the hopper and for equipment and components weighing over 25 kg. The lifting lugs shall be positioned to give optimum balance. Gross weight shall be marked in a conspicuous location and visible from lifting lug position.
- (ix) All manways shall be provided. Permanently installed davits are preferred for access covers. Hinges may be fitted where davits are not practical.
- (x) Manways shall be provided in any large division plates to permit access to all parts of the interior.
- (xi) A drain shall be fitted in the tank base to prevent liquid build up.
- (b) Hopper supports and associated support system:
 - (i) Supports system for the hopper systems which include slide gates, conveyors and appurtenances shall be designed and provided by the Contractor. Supports shall meet all specified design requirements and the general specification requirement. The location of the Hopper and the support system as shown in the drawings are indicative only and the Contractor shall field verify conditions and make all measurements and adjustments required.
 - (ii) No attempt has been made to show all required support locations, either on the Drawings or in the details. Support locations are only shown when the building frame requires that the support points be limited to specific locations.
 - (iii) The absence of defined support locations or support details on any Drawings shall not relieve the Contractor of the responsibility of providing all required hopper system supports throughout the facility. Supports shall not restrict access to other process systems.
 - (iv) Where discharge to road vehicle or skip is specified, the support structure shall be designed to permit the vehicle or skip to pass under the discharge point. The support structure shall be protected against vehicle impact. The controls shall be so located that operating staff are protected from vehicles.
 - (v) The supports of the hopper shall be designed in conformance with the AISC Manual of Steel Construction, and AISC Specification for the Design, Fabrication and Erection of Structural Steel for Buildings.
 - (vi) Hopper supports design shall be endorsed by a P.E. of the relevant discipline.
 - (vii) The maximum total deflection shall be limited to less than 1/300 of span.
 - (viii) Hopper supports shall be adequately braced to carry all horizontal and seismic loads.
 - (ix) The flange, webs, legs, or wall thickness of support framing shall not be less than 6 mm. Use tubular steel, pipe, flat bar or other members that provide a clean design which will minimise dust ledges and pockets.

- (x) The supports shall be shop fabricated of structural carbon steel conforming to ASTM A36. Steel plate shall conform to ASTM A36, or A283. Joining bolts shall conform to ASTM A325. All bolts and appurtenances necessary for erection and integration of the hopper support system and the hopper shall be supplied by the Contractor. All bolts shall be of 316L stainless steel.
- (c) Weigh Elements and Transmitters:
 - (i) The weight of hopper contents shall be indicated by means of compression type load cells which shall comply with the ICA specification.
 - (ii) A separate load cell shall be fitted to each leg. The readings shall be averaged to eliminate the effect of wind loading.
 - (iii) Weighing system for each hopper shall consist of four weigh elements and a weight indicating transmitter along with a panel mounted digital loop powered indicator.
 - (iv) Local and remote digital indicators shall be provided.
 - (v) Supplier shall verify required weigh element capacity and provide larger capacity elements if required.
 - (vi) Dummy cells shall be fitted during erection, and replaced by functional cells at the time of final commissioning.
 - (vii) The weigh elements shall be designed and installed so that the elements can be maintained or replaced.
 - (viii) All instrument shall comply with the ICA general specification.
 - (ix) The weigh element sensors shall produce an output voltage which is proportional to the weight. The weigh elements shall provide full temperature compensation and have an accuracy of 0.25 percent of rated capacity with a repeatability of 0.10 percent of rated capacity.
 - (x) The weight indicating transmitter shall be housed in an IP 66 enclosure, Type 316 stainless steel.
 - (xi) The weight indicating transmitter shall have a digital display that indicates net, gross, and tare weight. The display shall have six digits, 10 mm characters with the range of 0 to 999,999 with two fixed zeros.
 - (xii) Installation and calibration of the weigh elements and transmitter shall be the responsibility of the Contractor.
 - (xiii) The installation and calibration of the weighing system shall be checked by a factory representative of the weigh element supplier before start-up.

END OF SECTION